Runs on every example: figures by run and model

Runs of `docs/paper_plan.md`, made on 2026-10-11 on three decision models: Jev, Liquid d1 and GPT-6 Luna Decisions. The conclusions come first, then one page per finding with a chart, the evidence and one example, then the terms, and then one table per run with the models as rows. `python scripts/runs_report.py paper_runs --out docs/paper_runs.html` rebuilds this document from the results folders and `src/grjev/definitions/paper_runs.json`. The conclusions and their strength are Claude Code's reading of the figures on 2026-10-11, and Rushab has not reviewed them.

Conclusions

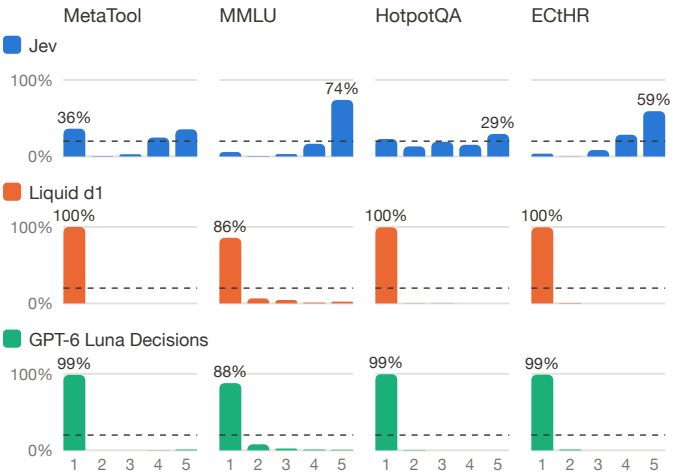
1. The probabilities of a list answer rank the options well and do not say whether an option is correct. The prediction is a correct option in 0.82 to 0.98 of the answers to lists with several correct options, and the probability of a correct option still goes from 0.11 to 0.85 with its place in the list and with the other options.
2. Order. Among 5 identical options, Liquid d1 and GPT-6 Luna Decisions predict the first one in 86% to 100% of the answers on all four datasets. Jev predicts the last one in 74% of the answers on MMLU and 59% on ECtHR. Strong.
3. Allocation. In a list with several correct options, one of them gets a mean probability of 0.66 to 0.85 and the lowest one 0.11 to 0.19, on three datasets and three models. Strong as a measurement.
4. Abstention. On lists with no correct option, Jev and Liquid d1 answer every list with a mean confidence of 0.49 to 0.87. GPT-6 Luna Decisions refuses 58% to 98% of the lists of random options and 1% to 27% of the lists of distractors of the example. Strong for random options.
5. Dependence on the list. When the correct option with the highest probability is removed, the mean probability of the second correct option goes from 0.11 to 0.19 up to 0.53 to 0.71. Strong.
6. Calibration. With one correct option the mean confidence is within 0.14 of the accuracy. It is 0.08 to 0.23 below the accuracy with several correct options. Weak as a finding of its own: it follows from the two findings above.
7. For the paper: findings on order, dependence on the list and abstention carry it, and each holds on four datasets or three. Before a submission the figures need intervals, a second run of the same requests, a check of the labels named under the limits, and the related work on option order and on questions with several correct answers.

1. Order: the position decides among identical options

A list of 5 copies of one correct option has no better option, and each model still prefers a position. Liquid d1 and GPT-6 Luna Decisions prefer the first position on every dataset. Jev prefers the last position on MMLU and ECtHR, the first and the last on MetaTool, and is near the equal share on HotpotQA.

Strength for the paper: strong for Liquid d1 and GPT-6 Luna Decisions. For Jev the bias exists and its direction changes with the dataset, so the claim for Jev is that the answer depends on the position, not that one position wins.

Share of the predictions at each position, 5 duplicates of one correct option



Evidence

- Predictions at position 1: Liquid d1 99.9%, 85.7%, 99.5% and 99.6% on MetaTool, MMLU, HotpotQA and ECtHR. GPT-6 Luna Decisions 98.6%, 87.8%, 99.4% and 98.9%. An equal share is 20%.
- Jev at the last position: 73.9% on MMLU, 59.3% on ECtHR, 35.5% on MetaTool, 29.5% on HotpotQA.
- The copy with the highest probability gets a mean of 0.50 to 0.81 on Liquid d1 and 0.67 to 0.79 on GPT-6 Luna Decisions. Equal probabilities are 0.20. Jev gives it 0.29 to 0.34.
- 5 paraphrases of one MetaTool tool: position 1 is predicted in 74% of the answers on Liquid d1 and 87% on GPT-6 Luna Decisions, and 26% on Jev.
- With one correct tool among distractors, the accuracy of GPT-6 Luna Decisions falls from 0.80 with the tool at the first place to 0.67 at the last. Jev goes from 0.78 to 0.80 and Liquid d1 from 0.69 to 0.74.
- 30,695 lists per model on four datasets, each list in its 5 cyclic permutations.

Limits

- The copies have names that differ by extra spaces, which the API needs to tell them apart.
- A list of identical options is not a list a user sends. The run with one correct tool among distractors is the link to real lists, and there the effect is 0.13 for one model and 0.05 or less for the two others.
- Each request was sent once. Jev does not return the same answer to the same request every time.

One example

The MMLU question, among the 1,000, whose probabilities by position are closest to the means of the three models.

Query. Which of the following groups or organizations can be identified as having the characteristics of a bureaucracy?

Instruction. Choose the correct answer to the question.

The list as sent: 5 copies of the correct choice

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**The government of a large city**	0.12	**0.48**	**0.74**
2	**The government of a large city**	0.10	0.13	0.16
3	**The government of a large city**	0.19	0.11	0.04
4	**The government of a large city**	0.26	0.11	0.04
5	**The government of a large city**	**0.33**	0.17	0.02

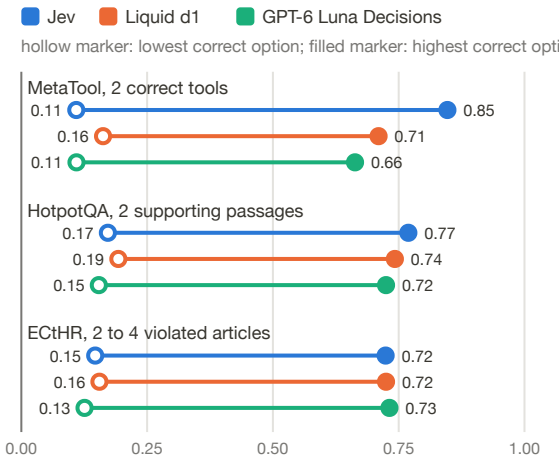
`mmlu/test/13562` in `results/mmlu_copied/2026-10-11_01`

2. Allocation: one of several correct options takes most of the probability

When a list holds 2 to 4 correct options, the models put most of the probability on one of them. The other correct options get about as much as a distractor would need to pass them. The wording of the instruction changes this by 0.01 to 0.02.

Strength for the paper: strong as a measurement: the same picture on three datasets and three models. The step from the measurement to a failure needs an argument. The probabilities of one answer sum to 1, so two correct options cannot both be near 1, and the instruction is the only place that says several options are wanted.

Mean probability of the lowest and of the highest correct option



Evidence

- Highest and lowest correct option, mean probability: 0.85 and 0.11 (Jev), 0.71 and 0.16 (Liquid d1), 0.66 and 0.11 (GPT-6 Luna Decisions) on MetaTool. 0.77 and 0.17, 0.74 and 0.19, 0.72 and 0.15 on HotpotQA. 0.72 and 0.15, 0.72 and 0.16, 0.73 and 0.13 on ECtHR.
- The lowest correct option is 0.05 to 0.14 above the highest distractor on average, and 0.06 below it for GPT-6 Luna Decisions on MetaTool. A threshold on the probability does not separate the second correct option from the distractors.
- The ranking is mostly right: the prediction is a correct option in 0.82 to 0.98 of the answers, and the 2 highest options are the 2 correct ones in 0.52 to 0.85 of them.
- Wording: "Choose one of them." and "Choose both." give the lowest correct MetaTool tool 0.14 and 0.15 on Jev. "may support", "supports" and "definitely supports" give the lowest supporting HotpotQA passage 0.18, 0.17 and 0.17.
- 7,405 HotpotQA questions, 497 MetaTool queries and 192 ECtHR cases.

Limits

- No reference says which probabilities are right for two correct options. 0.50 each is one reading of the instruction and not a fact about the model.
- A HotpotQA label marks a passage that holds a supporting sentence. Such a passage can be needed for the answer without supporting it alone, as the yes/no questions of finding 5 show.
- The ECtHR labels are the findings of the Court, not every article that the facts concern.

One example

The HotpotQA question closest to the means of the three models, among the questions with 10 passages in which the three models rank the same supporting passage first.

Query. Who is younger, Tony Roche or Renáta Tomanová?

Instruction. Choose every passage in the list that supports the answer to the question.

The 10 passages of the question

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**1976 Marlboro NSW Open** The 1976 New South Wales Open, also known by its sponsored name Marlboro New South W…	0.00	0.00	0.07
2	**Pavel Vojtisek** Paul Vojtischek (born Pavel Vojtíšek, 13 June 1963) is a Czechoslovak-born West Germ…	0.01	0.01	0.01
3	**1976 Australian Open – Women's Singles** Evonne Goolagong defeated Renáta Tomanová 6–2, 6–2 in the final to win the Women's S…	0.00	0.00	0.07
4	**Renáta Tomanová** correct Renáta Tomanová (born 9 December 1954) is a former professional tennis player from C…	**0.81**	**0.79**	**0.76**
5	**1980 French Open – Mixed Doubles** The Mixed Doubles tournament at the 1980 French Open was held from 26 May to 8 June…	0.00	0.00	0.00
6	**1979 Australian Open – Women's Doubles** Betsy Nagelsen and Renáta Tomanová were the defending champions.	0.00	0.00	0.01
7	**Tony Roche** correct Anthony Dalton Roche, AO MBE (born 17 May 1945) is a former professional Australian…	0.18	0.20	0.08
8	**1974 Australian Indoor Championships – Doubles** Rod Laver and John Newcombe were the defending champions but only Newcombe competed…	0.00	0.00	0.00
9	**Olga Zaitseva (tennis)** Olga Zaitseva (born 12 March 1962) is a former competitor for the Soviet Union at th…	0.00	0.00	0.00
10	**1976 French Open – Women's Singles** Two time reigning champion Chris Evert did not defend her title. She would stay home…	0.00	0.00	0.00

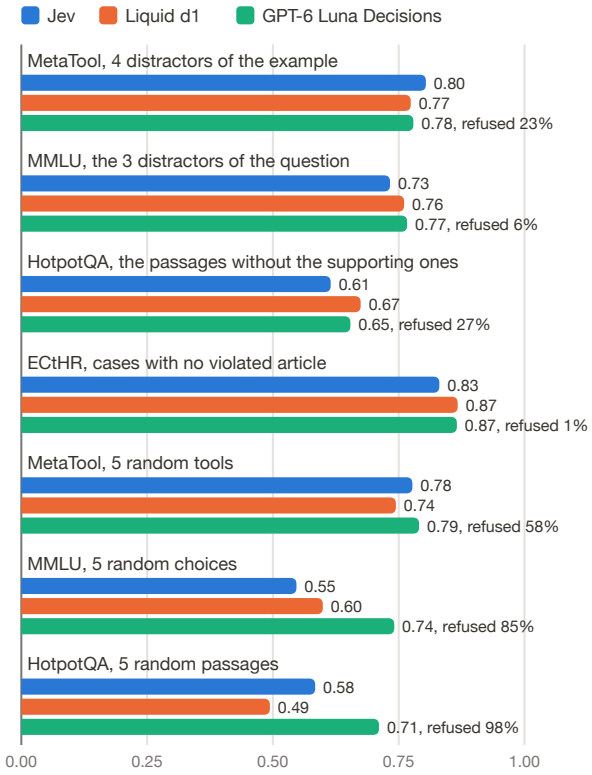
`hotpotqa/validation/5a7dcdea5542990b8f503abc` in `results/hotpotqa_removal/2026-10-11_01`

3. Abstention: confident answers to lists with no correct option

Jev and Liquid d1 answer every list that has no correct option, and their confidence there is close to their confidence on lists that have one. GPT-6 Luna Decisions declines many of these lists with an answer of type `refusal`, and is as confident as the two others on the lists that it answers. When NOTA is offered, the three models select it for random options and less often for the distractors of the example.

Strength for the paper: strong for lists of random options, where no option can be correct. Moderate for the distractors of the example, where the label decides that no option is correct. The refusals of GPT-6 Luna Decisions are a finding that the vendor does not document.

Mean confidence on lists with no correct option and no NOTA



One example

The MetaTool query closest to the mean confidence of Jev and Liquid d1, among the queries whose first list GPT-6 Luna Decisions refused.

Query. My child is interested in space, can you suggest any children's picture books about space exploration?

Instruction. Pick all the tools in the list that are relevant to the task at hand.

5 tools from outside the tool list of the query

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**SSH** Ability to SSH into your server and turn your natural language into server commands.	0.12	**0.66**	refused
2	**WeatherTool** Provide you with the latest weather information.	0.14	0.09	refused
3	**GifApi** Search through a wide range of gifs - Powered by Giphy.	**0.70**	0.21	refused
4	**airqualityforeast** Planning something outdoors? Get the 2-day air quality forecast for any US zip code.	0.02	0.02	refused
5	**haulingbuddies** Locate dependable animal transporters using recommendations, reviews, and regulatory…	0.02	0.02	refused

`metatool/scenario/1797` in `results/metatool_unrelated/2026-10-11_01`

Evidence

- Mean confidence on 5 random options: 0.78, 0.55 and 0.58 on Jev and 0.74, 0.60 and 0.49 on Liquid d1, for MetaTool, MMLU and HotpotQA.
- Mean confidence on the distractors of the example: 0.61 to 0.83 on Jev and 0.67 to 0.87 on Liquid d1, on four datasets. On lists with one correct option it is 0.86 to 0.94 and 0.81 to 0.89.
- GPT-6 Luna Decisions refuses 58%, 85% and 98% of the lists of random options, and 23%, 6%, 27% and 1% of the lists of distractors of the example. Jev and Liquid d1 refuse none of 202,088 questions each.
- With NOTA and random options, NOTA is the prediction in 0.93 to 1.00 of the answers. With NOTA and the distractors of the example it is 0.54 to 0.72 on MMLU and 0.79 to 0.88 on HotpotQA. For ECtHR cases with no violated article it is 0.37, 0.07 and 0.17.
- A request with one refused list was sent again 7 times with one change each: the same passages are answered with a query that one of them fits, and the same query is answered once NOTA or a fitting passage is in the list.

Limits

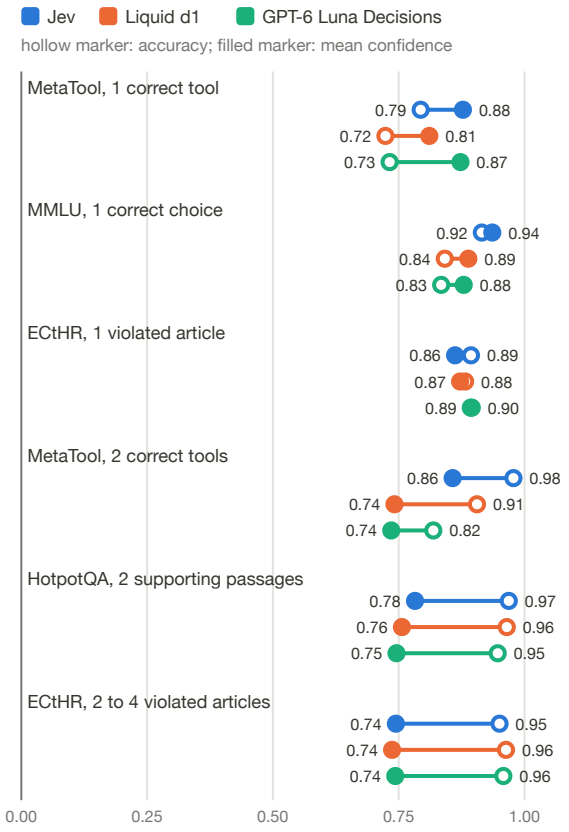
- The distractors of a MetaTool example can hold a tool that fits the query: the label of that run was read for 50 queries only.
- An ECtHR case with no label is a case in which the Court found no violation. Its facts can still point at an article.
- The figures of GPT-6 Luna Decisions are over the lists that it answers, 2% to 99% of them.
- No source states why the model refuses. The API returns no reason.

4. Calibration: close with one correct option, and it does not carry over

On lists with one correct option the confidence of the three models is close to their accuracy. The same models are 0.08 to 0.23 under their accuracy when the list holds several correct options, and their confidence is 0.49 to 0.87 when it holds none. A confidence threshold set on one kind of list does not hold on another.

Strength for the paper: weak as a finding of its own. The figures are sound, and the gaps follow from findings 2 and 3: a shared probability lowers the confidence, and a list with no correct option has accuracy 0. It supports the paper as a consequence.

Accuracy and mean confidence



Evidence

- One correct option, mean confidence minus accuracy: +0.08, +0.09 and +0.14 on MetaTool for Jev, Liquid d1 and GPT-6 Luna Decisions. +0.02, +0.05 and +0.04 on MMLU. -0.03, -0.01 and 0.00 on ECtHR.
- Several correct options: -0.12, -0.16 and -0.08 on MetaTool. -0.19, -0.21 and -0.20 on HotpotQA. -0.21, -0.23 and -0.21 on ECtHR.
- ECE with 10 bins is 0.01 to 0.15 with one correct option.
- 16,770 MetaTool lists, 4,000 MMLU lists and 6,550 ECtHR lists with one correct option.

Limits

- Accuracy counts a prediction as correct when it is one of the correct options. With several correct options another definition gives another gap.
- MetaTool labels one tool per query, and the three models agree on another answer in some queries, as in the example.
- No interval is computed yet.

One example

Among the MetaTool queries on which Jev is wrong with a confidence of 0.90 or more on the released order, the one with the shortest list and then the lowest id.

Query. Yes, I can definitely assist you with that. Could you please specify whether you prefer a step-by-step explanation or would you like me to provide you with a sample code snippet as well?

Instruction. Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None.

The tool list of the query and "None", in the released order

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**RepoTool** correct Discover GitHub projects tailored to your needs, explore their structures with insig…	0.02	0.01	0.00
2	**AI2sql** Converts a natural language text into an SQL query.	0.00	0.00	0.00
3	**SSH** Ability to SSH into your server and turn your natural language into server commands.	0.00	0.00	0.00
4	**AutoInfra1** Talk to your Servers. Works with AWS, GCP, Azure, and anywhere you can ssh!	0.00	0.00	0.00
5	**noteable** Create notebooks in Python, SQL, and Markdown to explore data, visualize, and share…	0.00	0.00	0.00
6	**dart** Project management on autopilot.	0.00	0.00	0.00
7	**hackit_web_scanner** AI Powered Web Scanner by HACKIT.	0.00	0.00	0.00
8	**LarkBaseImporter** Importing data into Lark Base for further analysis and presentation. An easy-to-use…	0.00	0.00	0.00
9	**webhooks** Allows you to write, deploy, and manage HTTP Webhooks in JavaScript, right from the…	0.00	0.00	0.00
10	**universal** Enables to access web pages, analyze PDFs, generate QR codes, etc.	0.00	0.00	0.00
11	**None**	**0.98**	**0.99**	**1.00**

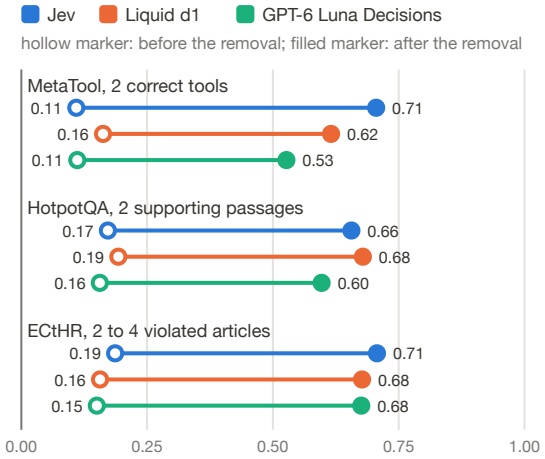
`metatool/scenario/0` in `results/metatool_position/2026-10-11_01`

5. Dependence on the list: the second correct option rises when the first is removed

The probability of an option is a share of the list and not a judgement of the option alone. When the correct option with the highest probability is taken out, the second correct option gets most of what it had.

Strength for the paper: strong: all nine cells of three datasets and three models move the same way by 0.42 to 0.60, on 7,384 HotpotQA questions, 497 MetaTool queries and 192 ECtHR cases.

Mean probability of the second correct option, before and after the removal



Evidence

- Second correct option before and after the removal: 0.11 to 0.71, 0.16 to 0.62 and 0.11 to 0.53 on MetaTool for Jev, Liquid d1 and GPT-6 Luna Decisions. 0.17 to 0.66, 0.19 to 0.68 and 0.16 to 0.60 on HotpotQA. 0.19 to 0.71, 0.16 to 0.68 and 0.15 to 0.68 on ECtHR.
- On the list without one correct option, the prediction is the remaining correct option in 0.86, 0.78 and 0.67 of the MetaTool answers and in 0.84, 0.84 and 0.80 of the HotpotQA answers.
- Asked alone in a yes/no question, a violated ECtHR article gets a mean probability of yes of 0.67, 0.89 and 0.74, and 80% to 96% of them get 0.50 or more.
- A supporting HotpotQA passage asked alone gets 0.45, 0.40 and 0.52, and 38% to 51% of them get 0.50 or more.

Limits

- A probability that sums to 1 over the list has to move when an option leaves. The finding is the size: the second correct option is not distinguishable from a distractor before the removal.
- The yes/no figures of HotpotQA say that about half of the supporting passages are not judged to support the answer alone. This weakens the reading that both passages are correct options there.
- The yes/no questions of HotpotQA cover 500 questions.

One example

The question of finding 2, after the removal.

Query. Who is younger, Tony Roche or Renáta Tomanová?

Instruction. Choose every passage in the list that supports the answer to the question.

The question of finding 2 without Renáta Tomanová, the supporting passage with the highest probability. In the full list Tony Roche has 0.18, 0.20 and 0.08.

#	Option	Jev	Liquid d1	GPT-6 Luna Decisions
1	**1976 Marlboro NSW Open** The 1976 New South Wales Open, also known by its sponsored name Marlboro New South W…	0.01	0.07	**0.44**
2	**Pavel Vojtisek** Paul Vojtischek (born Pavel Vojtíšek, 13 June 1963) is a Czechoslovak-born West Germ…	0.34	0.22	0.03
3	**1976 Australian Open – Women's Singles** Evonne Goolagong defeated Renáta Tomanová 6–2, 6–2 in the final to win the Women's S…	0.01	0.05	0.08
4	**1980 French Open – Mixed Doubles** The Mixed Doubles tournament at the 1980 French Open was held from 26 May to 8 June…	0.01	0.03	0.01
5	**1979 Australian Open – Women's Doubles** Betsy Nagelsen and Renáta Tomanová were the defending champions.	0.01	0.01	0.01
6	**Tony Roche** correct Anthony Dalton Roche, AO MBE (born 17 May 1945) is a former professional Australian…	**0.60**	**0.53**	0.41
7	**1974 Australian Indoor Championships – Doubles** Rod Laver and John Newcombe were the defending champions but only Newcombe competed…	0.00	0.00	0.00
8	**Olga Zaitseva (tennis)** Olga Zaitseva (born 12 March 1962) is a former competitor for the Soviet Union at th…	0.01	0.06	0.00
9	**1976 French Open – Women's Singles** Two time reigning champion Chris Evert did not defend her title. She would stay home…	0.01	0.01	0.02

`hotpotqa/validation/5a7dcdea5542990b8f503abc` in `results/hotpotqa_removal/2026-10-11_01`

Terms

- **Option**: one entry of the list sent with a query: a tool on MetaTool, a choice on MMLU, a passage on HotpotQA, an article on ECtHR.
- **Correct option**: an option that the dataset labels as correct for the query.
- **Distractor**: an option that is not a correct option.
- **NOTA**: an option that says no other option is correct: "None" on MetaTool, "None of the above" on MMLU, "None of the passages" on HotpotQA, "None of these articles" on ECtHR. It is the correct option of a list that holds no other correct option.
- **Prediction**: the option that the model selects, the field `choice` of its answer.
- **Confidence**: the highest probability of an answer. It is not the field `confidence` of the model's answer.
- **Refused list**: a list for which the model returns an answer of type `refusal` and no probabilities.
- **Accuracy**: the share of the answers whose prediction is a correct option.
- **Exact match**: for a list with k correct options, k being 2 or more: the share of the answers whose k options with the highest probabilities are the k correct options. A tie for place k is not a match.
- **Overconfidence**: mean confidence minus accuracy.
- **ECE (expected calibration error)**: the answers are put in 10 bins by confidence, from 0 to 0.1 up to 0.9 to 1.0. Each bin gives the distance between its mean confidence and its accuracy. ECE is the mean of the 10 distances, weighted by the number of answers in the bin.
- **Prediction minus runner-up**: the highest probability of an answer minus its second-highest probability.
- **Highest correct minus lowest correct**: the highest probability of a correct option minus the lowest probability of a correct option.
- **Highest correct minus highest distractor**: the highest probability of a correct option minus the highest probability of a distractor.
- **Lowest correct minus highest distractor**: the lowest probability of a correct option minus the highest probability of a distractor. It is negative when a distractor has a higher probability than a correct option.
- **Entropy**: the sum of -p log2(p) over the probabilities p of an answer, in bits. It is 0 when one option has 1.00. Its maximum is log2 of the number of options, reached when every option has the same probability, and the header of the column gives it.
- **Position**: the place of an option in the list sent, counted from the first.

Notes

- Jev is `jev-1.13.0`, asked through TypeSafe's API. Liquid d1 (`liquid/d1`) and GPT-6 Luna Decisions (`openai/gpt-6-luna-decisions`) are asked through Vercel AI Gateway, which takes no model version. The three models get the same requests.
- Every figure of a row is computed over the lists that the model answered. A refused list has no answer. Jev and Liquid d1 refused no list.
- A column is left out of a table when no model has that figure: a list with no correct option has no accuracy, and a list with only correct options has no distractor.
- The three differences and the probabilities are computed for each answer and then averaged over the answers. The differences with a distractor use the answers to lists that hold a correct option and a distractor. Where a list of a run holds only correct options, the column "Lists with a correct option and a distractor" gives their number.
- The MMLU runs use a seeded sample of 1,000 of the 13,096 questions whose choices can be moved. The HotpotQA runs with 5 cyclic permutations and the HotpotQA run with 4 distractors and NOTA use a seeded sample of 1,000 of the 7,345 questions with 10 passages. The other runs use every example.
- MetaTool repeats queries across its test files. The 1,767 examples of the run with paraphrases give 1,047 different requests, and the 3,292 examples of the run with duplicates 3,275. An answer to a repeated request is counted once per example.
- The labels of ECtHR are the articles that the Court found violated in a case, and a case with no label is one in which it found no violation of the 10 articles.
- The yes/no question of a HotpotQA passage is sent for 500 questions, and that of an ECtHR article for the 1,000 test cases.

Tables by run

1. Order: lists whose options are all correct

MetaTool: 5 duplicates of one correct tool

The list holds 5 copies of one correct tool of the example. The names of the copies differ only by extra spaces. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 3,292 examples, 16,460 lists. Options in a list: 5. Correct options in a list: 5.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	16,460	0	0.32	0.09	+0.21	0.321, 0.232, 0.186, 0.148, 0.112	2.21	0.23, 0.12, 0.17, 0.23, 0.25	5,956, 86, 494, 4,073, 5,851
Liquid d1	16,460	0	0.81	0.73	+0.78	0.806, 0.078, 0.054, 0.035, 0.026	1.03	0.81, 0.07, 0.03, 0.03, 0.06	16,445, 10, 0, 0, 5
GPT-6 Luna Decisions	16,198	262	0.78	0.66	+0.77	0.783, 0.127, 0.049, 0.025, 0.015	1.02	0.78, 0.10, 0.04, 0.03, 0.05	15,970, 5, 11, 43, 169

`results/metatool_copied/2026-10-11_01`, every list

MMLU: 5 duplicates of the correct choice

The list holds 5 copies of the correct choice of the question. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 5.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.33	0.09	+0.23	0.334, 0.240, 0.188, 0.136, 0.102	2.18	0.14, 0.12, 0.19, 0.24, 0.31	296, 27, 160, 824, 3,693
Liquid d1	5,000	0	0.50	0.31	+0.42	0.502, 0.187, 0.133, 0.102, 0.077	1.85	0.48, 0.16, 0.13, 0.10, 0.13	4,283, 329, 219, 52, 117
GPT-6 Luna Decisions	4,832	168	0.74	0.58	+0.73	0.743, 0.164, 0.053, 0.026, 0.014	1.03	0.71, 0.17, 0.06, 0.03, 0.03	4,242, 379, 116, 55, 40

`results/mmlu_copied/2026-10-11_01`, every list

HotpotQA: 5 duplicates of one supporting passage

The list holds 5 copies of one supporting passage of the question. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 5.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.34	0.11	+0.23	0.341, 0.226, 0.176, 0.143, 0.114	2.18	0.20, 0.18, 0.21, 0.19, 0.23	1,146, 664, 955, 761, 1,474
Liquid d1	5,000	0	0.73	0.63	+0.68	0.728, 0.101, 0.072, 0.057, 0.043	1.33	0.73, 0.09, 0.06, 0.05, 0.08	4,975, 10, 10, 0, 5
GPT-6 Luna Decisions	4,621	379	0.79	0.68	+0.78	0.792, 0.115, 0.049, 0.029, 0.015	1.00	0.79, 0.11, 0.04, 0.02, 0.05	4,595, 20, 3, 1, 2

`results/hotpotqa_copied/2026-10-11_01`, every list

ECtHR: 5 duplicates of one violated article

The list holds 5 copies of one article that the Court found violated in the case. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 847 examples, 4,235 lists. Options in a list: 5. Correct options in a list: 5.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	4,235	0	0.29	0.06	+0.17	0.295, 0.236, 0.194, 0.154, 0.122	2.24	0.16, 0.13, 0.21, 0.23, 0.27	153, 11, 357, 1,203, 2,511
Liquid d1	4,235	0	0.55	0.36	+0.48	0.547, 0.189, 0.109, 0.084, 0.070	1.80	0.55, 0.19, 0.10, 0.08, 0.08	4,217, 18, 0, 0, 0
GPT-6 Luna Decisions	4,179	56	0.67	0.45	+0.65	0.665, 0.212, 0.069, 0.033, 0.020	1.35	0.66, 0.21, 0.07, 0.03, 0.03	4,134, 41, 0, 4, 0

`results/ecthr_copied/2026-10-11_01`, every list

MetaTool: 5 paraphrases of one correct tool

The list holds 5 descriptions of one correct tool in 5 styles, written by Claude Opus 5.5. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,767 examples, 8,835 lists. Options in a list: 5. Correct options in a list: 5.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Highest correct minus lowest correct	Mean probability of each correct option, from the highest	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	8,835	0	0.47	0.23	+0.42	0.468, 0.242, 0.152, 0.090, 0.047	1.85	0.23, 0.18, 0.17, 0.20, 0.21	2,336, 1,518, 1,279, 1,713, 1,989
Liquid d1	8,835	0	0.52	0.30	+0.46	0.517, 0.214, 0.124, 0.085, 0.059	1.80	0.45, 0.16, 0.12, 0.11, 0.17	6,571, 454, 415, 408, 987
GPT-6 Luna Decisions	8,155	680	0.59	0.37	+0.56	0.590, 0.223, 0.102, 0.054, 0.031	1.56	0.56, 0.19, 0.09, 0.06, 0.09	7,124, 670, 164, 81, 116

`results/metatool_rotated/2026-10-11_01`, every list

MetaTool: one correct tool at the first place of the list

The tool list of the example with its correct tool moved to the first place. "None" is the last option. Test files `similar_tools` and `scenario`. 2,795 examples, 2,795 lists. Options in a list: 6 to 16. Correct options in a list: 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	2,795	0	0.78	0.87	+0.08	0.08	0.77	+0.53	0.744	0.216	0.48
Liquid d1	2,795	0	0.69	0.80	+0.10	0.11	0.65	+0.33	0.629	0.303	0.78
GPT-6 Luna Decisions	2,793	2	0.80	0.89	+0.09	0.09	0.80	+0.55	0.760	0.206	0.44

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, list `first`

MetaTool: one correct tool at the middle of the list

The tool list of the example with its correct tool moved to the middle. "None" is the last option. Test files `similar_tools` and `scenario`. 2,795 examples, 2,795 lists. Options in a list: 6 to 16. Correct options in a list: 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	2,795	0	0.80	0.88	+0.08	0.08	0.79	+0.57	0.765	0.200	0.42
Liquid d1	2,795	0	0.74	0.82	+0.08	0.09	0.69	+0.43	0.686	0.252	0.70
GPT-6 Luna Decisions	2,793	2	0.72	0.87	+0.15	0.15	0.76	+0.42	0.686	0.266	0.53

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, list `middle`

MetaTool: one correct tool at the last place of the list

The tool list of the example with its correct tool moved to the last place. "None" is the last option. Test files `similar_tools` and `scenario`. 2,795 examples, 2,795 lists. Options in a list: 6 to 16. Correct options in a list: 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	2,795	0	0.80	0.88	+0.08	0.08	0.79	+0.56	0.763	0.202	0.43
Liquid d1	2,795	0	0.74	0.83	+0.09	0.09	0.70	+0.45	0.696	0.245	0.66
GPT-6 Luna Decisions	2,793	2	0.67	0.86	+0.19	0.19	0.75	+0.35	0.647	0.301	0.53

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, list `last`

2. Allocation: lists with several correct options

MetaTool: two correct tools, "Choose one of them."

The tool list of the example, which holds two correct tools, in 9 orders: the released order, the two correct tools next to each other at 5 places, and apart at 3 pairs of places. 497 examples, 4,473 lists. Options in a list: 10. Correct options in a list: 2.

Instruction: "Two tools in the list are appropriate to solve the user's query. Choose one of them."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	4,473	0	0.97	0.74	0.82	-0.15	0.15	0.67	+0.68	+0.77	+0.09	0.817, 0.136	0.045	0.61
Liquid d1	4,473	0	0.95	0.72	0.77	-0.17	0.17	0.58	+0.60	+0.68	+0.08	0.758, 0.159	0.075	0.77
GPT-6 Luna Decisions	4,473	0	0.93	0.60	0.79	-0.14	0.14	0.64	+0.64	+0.66	+0.02	0.761, 0.118	0.097	0.81

`results/metatool_position/2026-10-11_02`, every list

MetaTool: two correct tools, "Choose both."

The tool list of the example, which holds two correct tools, in 9 orders: the released order, the two correct tools next to each other at 5 places, and apart at 3 pairs of places. 497 examples, 4,473 lists. Options in a list: 10. Correct options in a list: 2.

Instruction: "Two tools in the list are appropriate to solve the user's query. Choose both."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	4,473	0	0.98	0.82	0.82	-0.16	0.16	0.65	+0.66	+0.77	+0.11	0.809, 0.149	0.039	0.65
Liquid d1	4,473	0	0.95	0.74	0.75	-0.20	0.20	0.54	+0.56	+0.65	+0.09	0.737, 0.173	0.083	0.84
GPT-6 Luna Decisions	4,473	0	0.90	0.57	0.74	-0.16	0.16	0.58	+0.58	+0.57	-0.01	0.697, 0.116	0.123	1.15

`results/metatool_position/2026-10-11_03`, every list

MetaTool: two correct tools, every needed tool is asked for

The tool list of the example in the released order. 497 examples, 497 lists. Options in a list: 10. Correct options in a list: 2.

Instruction: "Choose every tool in the list that is needed to solve the user's query."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	497	0	0.98	0.73	0.86	-0.12	0.12	0.73	+0.74	+0.81	+0.07	0.847, 0.109	0.042	0.52
Liquid d1	497	0	0.91	0.70	0.74	-0.16	0.17	0.53	+0.55	+0.59	+0.05	0.710, 0.162	0.117	0.88
GPT-6 Luna Decisions	497	0	0.82	0.52	0.74	-0.08	0.08	0.56	+0.55	+0.49	-0.06	0.663, 0.110	0.173	1.09

`results/metatool_removal/2026-10-11_01`, list `full`

HotpotQA: the passages of the question, "supports"

The passages of the question in the released order. Two of them hold a supporting fact. 7,405 examples, 7,405 lists. Options in a list: 2 to 10. Correct options in a list: 2.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Lists with a correct option and a distractor	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.32)
Jev	7,405	0	7,384	0.97	0.82	0.78	-0.19	0.19	0.60	+0.60	+0.72	+0.13	0.769, 0.172	0.046	0.74
Liquid d1	7,405	0	7,384	0.96	0.85	0.76	-0.21	0.21	0.55	+0.55	+0.69	+0.14	0.743, 0.193	0.050	0.83
GPT-6 Luna Decisions	7,396	9	7,375	0.95	0.70	0.75	-0.20	0.20	0.57	+0.57	+0.65	+0.08	0.725, 0.154	0.077	1.00

`results/hotpotqa_removal/2026-10-11_01`, list `full`

HotpotQA: the passages of the question, "may support"

The passages of the question in the released order. Two of them hold a supporting fact. 7,405 examples, 7,405 lists. Options in a list: 2 to 10. Correct options in a list: 2.

Instruction: "Choose every passage in the list that may support the answer to the question."

Model	Lists answered	Lists refused	Lists with a correct option and a distractor	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.32)
Jev	7,405	0	7,384	0.97	0.82	0.78	-0.19	0.19	0.58	+0.59	+0.72	+0.13	0.765, 0.179	0.044	0.75
Liquid d1	7,405	0	7,384	0.96	0.85	0.76	-0.21	0.21	0.55	+0.55	+0.69	+0.14	0.742, 0.192	0.051	0.85
GPT-6 Luna Decisions	7,401	4	7,380	0.94	0.70	0.75	-0.20	0.20	0.57	+0.57	+0.65	+0.08	0.725, 0.155	0.078	0.99

`results/hotpotqa_removal/2026-10-11_01`, list `full_may`

HotpotQA: the passages of the question, "definitely supports"

The passages of the question in the released order. Two of them hold a supporting fact. 7,405 examples, 7,405 lists. Options in a list: 2 to 10. Correct options in a list: 2.

Instruction: "Choose every passage in the list that definitely supports the answer to the question."

Model	Lists answered	Lists refused	Lists with a correct option and a distractor	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.32)
Jev	7,405	0	7,384	0.97	0.81	0.79	-0.18	0.18	0.60	+0.60	+0.73	+0.12	0.772, 0.169	0.045	0.73
Liquid d1	7,405	0	7,384	0.96	0.85	0.76	-0.21	0.21	0.55	+0.55	+0.69	+0.14	0.743, 0.192	0.050	0.83
GPT-6 Luna Decisions	7,387	18	7,366	0.95	0.68	0.74	-0.20	0.20	0.57	+0.57	+0.64	+0.07	0.723, 0.151	0.079	1.02

`results/hotpotqa_removal/2026-10-11_01`, list `full_definitely`

ECtHR: cases with 2 to 4 violated articles

The 10 articles. Each example is sent with the 10 cyclic permutations of the list, so each option is at each position once. 192 examples, 1,920 lists. Options in a list: 10. Correct options in a list: 2 to 4.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	1,920	0	0.95	0.76	0.74	-0.21	0.21	0.54	+0.58	+0.65	+0.08	highest 0.724, lowest 0.147	0.069	0.89
Liquid d1	1,920	0	0.96	0.73	0.74	-0.23	0.23	0.54	+0.57	+0.66	+0.09	highest 0.725, lowest 0.155	0.065	0.96
GPT-6 Luna Decisions	1,920	0	0.96	0.70	0.74	-0.21	0.21	0.57	+0.61	+0.67	+0.07	highest 0.732, lowest 0.125	0.060	1.06

`results/ecthr_own/2026-10-11_01`, cases with 2 to 4 labels, every list

3. Abstention: lists with no correct option

MetaTool: 5 random tools, no "None"

The list holds 5 tools of the dataset that are not in the tool list of the example, chosen with a seed. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 3,292 examples, 16,460 lists. Options in a list: 5. Correct options in a list: 0.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	16,460	0	0.78	0.63	0.83	0.20, 0.21, 0.18, 0.20, 0.20	3,405, 3,457, 2,991, 3,328, 3,279
Liquid d1	16,460	0	0.74	0.60	0.98	0.19, 0.21, 0.21, 0.18, 0.22	3,260, 3,458, 3,383, 2,851, 3,508
GPT-6 Luna Decisions	6,950	9,510	0.79	0.67	0.94	0.25, 0.21, 0.19, 0.16, 0.19	1,568, 1,466, 1,313, 1,151, 1,452

`results/metatool_unrelated/2026-10-11_01`, every list

MMLU: 5 random choices, no NOTA

The list holds 5 choices of other questions, chosen with a seed. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 0.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.55	0.34	1.66	0.22, 0.22, 0.17, 0.20, 0.20	1,083, 1,115, 768, 1,028, 1,006
Liquid d1	5,000	0	0.60	0.41	1.48	0.28, 0.24, 0.19, 0.12, 0.17	1,580, 1,258, 931, 467, 764
GPT-6 Luna Decisions	772	4,228	0.74	0.61	1.11	0.38, 0.23, 0.14, 0.11, 0.14	334, 179, 93, 78, 88

`results/mmlu_unrelated/2026-10-11_01`, every list

HotpotQA: 5 random passages, no NOTA

The list holds 5 passages of other questions, chosen with a seed. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 0.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.58	0.38	1.55	0.21, 0.22, 0.18, 0.19, 0.20	1,012, 1,186, 881, 925, 996
Liquid d1	5,000	0	0.49	0.29	1.78	0.14, 0.25, 0.22, 0.16, 0.23	488, 1,473, 1,105, 637, 1,297
GPT-6 Luna Decisions	99	4,901	0.71	0.55	1.25	0.22, 0.18, 0.19, 0.09, 0.32	16, 19, 17, 10, 37

`results/hotpotqa_unrelated/2026-10-11_01`, every list

MetaTool: 4 distractors of the example, no "None"

The list holds 4 tools of the tool list of the example that are not correct. Each example is sent with the 4 cyclic permutations of the list, so each option is at each position once. 995 examples, 3,980 lists. Options in a list: 4. Correct options in a list: 0.

Instruction: "Pick all the tools in the list that are relevant to the task at hand."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.00)	Mean probability at each position	Predictions at each position
Jev	3,980	0	0.80	0.65	0.68	0.26, 0.26, 0.24, 0.25	1,036, 1,017, 939, 988
Liquid d1	3,980	0	0.77	0.61	0.82	0.27, 0.26, 0.25, 0.22	1,117, 1,023, 991, 849
GPT-6 Luna Decisions	3,059	921	0.78	0.63	0.86	0.29, 0.27, 0.24, 0.20	916, 826, 711, 606

`results/metatool_distractors/2026-10-11_01`, every list

MMLU: the 3 distractors of the question, no NOTA

The list holds the 3 choices of the question that are not correct. Each example is sent with the 3 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 3,000 lists. Options in a list: 3. Correct options in a list: 0.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 1.58)	Mean probability at each position	Predictions at each position
Jev	3,000	0	0.73	0.53	0.89	0.35, 0.34, 0.32	1,059, 1,018, 923
Liquid d1	3,000	0	0.76	0.58	0.82	0.33, 0.34, 0.33	1,013, 1,025, 962
GPT-6 Luna Decisions	2,832	168	0.77	0.59	0.80	0.42, 0.30, 0.29	1,246, 796, 790

`results/mmlu_distractors/2026-10-11_01`, every list

MetaTool: the tool list without its two correct tools

The tool list of the example in the released order, without the two correct tools. 497 examples, 497 lists. Options in a list: 8. Correct options in a list: 0.

Instruction: "Choose every tool in the list that is needed to solve the user's query."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 3.00)
Jev	497	0	0.82	0.68	0.67
Liquid d1	497	0	0.79	0.67	0.85
GPT-6 Luna Decisions	405	92	0.76	0.62	1.06

`results/metatool_removal/2026-10-11_01`, list `without_all`

HotpotQA: the passages without the supporting passages

The passages of the question in the released order, without the passages that hold a supporting fact. 7,377 examples, 7,377 lists. Options in a list: 2 to 8. Correct options in a list: 0.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 1.00 to 3.00)
Jev	7,377	0	0.61	0.43	1.51
Liquid d1	7,377	0	0.67	0.51	1.32
GPT-6 Luna Decisions	5,352	2,025	0.65	0.50	1.48

`results/hotpotqa_removal/2026-10-11_01`, list `without_all`

ECtHR: the articles without the violated articles

The 10 articles in the released order, without the articles that the Court found violated. 847 examples, 847 lists. Options in a list: 6 to 9. Correct options in a list: 0.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 2.58 to 3.17)
Jev	847	0	0.74	0.57	0.99
Liquid d1	847	0	0.74	0.60	1.09
GPT-6 Luna Decisions	610	237	0.71	0.58	1.39

`results/ecthr_removal/2026-10-11_01`, list `without_all`

ECtHR: cases with no violated article, no NOTA

The 10 articles. Each example is sent with the 10 cyclic permutations of the list, so each option is at each position once. 153 examples, 1,530 lists. Options in a list: 10. Correct options in a list: 0.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Mean confidence	Prediction minus runner-up	Mean entropy in bits (maximum 3.32)
Jev	1,530	0	0.83	0.71	0.66
Liquid d1	1,530	0	0.87	0.77	0.54
GPT-6 Luna Decisions	1,515	15	0.87	0.78	0.62

`results/ecthr_own/2026-10-11_01`, cases with no label, every list

MetaTool: 4 random tools and "None"

The list holds 4 tools of the dataset that are not in the tool list of the example, and "None", which is the correct option. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 995 examples, 4,975 lists. Options in a list: 5. Correct options in a list: 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	4,975	0	0.95	0.95	0.00	0.01	0.91	+0.86	0.929	0.068	0.16	0.20, 0.20, 0.20, 0.20, 0.20	990, 1,009, 972, 1,002, 1,002
Liquid d1	4,975	0	0.98	0.97	-0.01	0.01	0.93	+0.92	0.957	0.040	0.14	0.20, 0.20, 0.20, 0.20, 0.20	985, 995, 999, 990, 1,006
GPT-6 Luna Decisions	4,966	9	0.93	0.95	+0.02	0.02	0.91	+0.82	0.902	0.087	0.24	0.20, 0.20, 0.20, 0.20, 0.19	1,014, 999, 987, 987, 979

`results/metatool_absent_random/2026-10-11_01`, every list

MMLU: 4 random choices and NOTA

The list holds 4 choices of other questions and NOTA, which is the correct option. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 1.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.97	0.95	-0.03	0.03	0.91	+0.89	0.936	0.049	0.26	0.20, 0.20, 0.20, 0.20, 0.20	1,006, 998, 987, 1,001, 1,008
Liquid d1	5,000	0	0.99	0.98	-0.01	0.01	0.96	+0.95	0.973	0.021	0.10	0.20, 0.20, 0.20, 0.20, 0.20	1,005, 997, 1,001, 998, 999
GPT-6 Luna Decisions	4,996	4	0.99	0.94	-0.05	0.05	0.90	+0.90	0.936	0.041	0.34	0.21, 0.21, 0.20, 0.19, 0.19	1,007, 1,002, 996, 995, 996

`results/mmlu_absent_random/2026-10-11_01`, every list

HotpotQA: 4 random passages and NOTA

The list holds 4 passages of other questions and NOTA, which is the correct option. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 1.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each positior
Jev	5,000	0	1.00	1.00	0.00	0.00	0.99	+0.99	0.995	0.004	0.03	0.20, 0.20, 0.20, 0.20, 0.20	1,000, 1,000, 1,000, 1,000, 1,000
Liquid d1	5,000	0	1.00	1.00	0.00	0.00	1.00	+1.00	1.000	0.000	0.00	0.20, 0.20, 0.20, 0.20, 0.20	1,000, 1,000, 1,000, 1,000, 1,000
GPT-6 Luna Decisions	5,000	0	1.00	0.94	-0.06	0.06	0.91	+0.91	0.943	0.036	0.34	0.22, 0.19, 0.20, 0.20, 0.20	1,001, 1,000, 1,000, 999, 1,000

`results/hotpotqa_absent_random/2026-10-11_01`, every list

MMLU: the 3 distractors of the question and NOTA

The list holds the 3 choices of the question that are not correct and NOTA, which is the correct option. Each example is sent with the 4 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 4,000 lists. Options in a list: 4. Correct options in a list: 1.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.00)	Mean probability at each position	Predictions at each position
Jev	4,000	0	0.65	0.77	+0.12	0.12	0.61	+0.23	0.575	0.341	0.85	0.26, 0.25, 0.24, 0.25	1,053, 998, 971, 978
Liquid d1	4,000	0	0.72	0.81	+0.10	0.10	0.68	+0.40	0.668	0.273	0.69	0.22, 0.25, 0.27, 0.26	880, 978, 1,097, 1,045
GPT-6 Luna Decisions	4,000	0	0.54	0.76	+0.22	0.22	0.59	+0.10	0.497	0.395	0.89	0.28, 0.25, 0.25, 0.23	1,183, 963, 940, 914

`results/mmlu_absent/2026-10-11_01`, every list

HotpotQA: 4 distractors of the question and NOTA

The list holds 4 passages of the question that hold no supporting fact and NOTA, which is the correct option. Each example is sent with the 5 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 5,000 lists. Options in a list: 5. Correct options in a list: 1.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.32)	Mean probability at each position	Predictions at each position
Jev	5,000	0	0.84	0.82	-0.01	0.03	0.69	+0.55	0.752	0.199	0.67	0.19, 0.21, 0.19, 0.20, 0.21	973, 1,047, 970, 998, 1,012
Liquid d1	5,000	0	0.88	0.90	+0.02	0.02	0.82	+0.70	0.840	0.136	0.39	0.19, 0.20, 0.20, 0.20, 0.21	975, 999, 1,015, 992, 1,019
GPT-6 Luna Decisions	5,000	0	0.79	0.77	-0.01	0.04	0.63	+0.44	0.677	0.232	0.92	0.24, 0.20, 0.20, 0.18, 0.18	1,184, 967, 998, 905, 946

`results/hotpotqa_absent/2026-10-11_01`, every list

MetaTool `reliability`: the tool list of the example and "None"

The tool list of the example in the released order with "None" as the last option. No tool of the list is correct, and "None" is the correct option. 995 examples, 995 lists. Options in a list: 11. Correct options in a list: 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.46)
Jev	995	0	0.91	0.91	+0.01	0.02	0.84	+0.74	0.864	0.127	0.31
Liquid d1	995	0	0.98	0.94	-0.03	0.03	0.90	+0.88	0.933	0.056	0.25
GPT-6 Luna Decisions	993	2	0.94	0.95	+0.01	0.03	0.92	+0.84	0.912	0.077	0.21

`results/metatool_position/2026-10-11_01`, test file `reliability`, list `released`

ECtHR: the 10 articles and NOTA, cases with no violated article

The 10 articles in the released order with NOTA as the last option, which is the correct option. 153 examples, 153 lists. Options in a list: 11. Correct options in a list: 1.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.46)
Jev	153	0	0.37	0.66	+0.29	0.30	0.39	-0.13	0.392	0.524	1.09
Liquid d1	153	0	0.07	0.79	+0.72	0.72	0.64	-0.64	0.132	0.767	0.81
GPT-6 Luna Decisions	153	0	0.17	0.72	+0.55	0.55	0.54	-0.42	0.233	0.652	1.09

`results/ecthr_with_none/2026-10-11_01`, cases with no label, list `own`

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4. Calibration: lists with one correct option

MetaTool: one correct tool, the 6 orders of the list

The tool list of the example in the released order and with its correct tool at 5 places. "None" is the last option. Test files `similar_tools` and `scenario`. 2,795 examples, 16,770 lists. Options in a list: 6 to 16. Correct options in a list: 1.

Instruction: "Choose the tool that is applicable to the user's query. If no tool in the list is applicable, choose None."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.58 to 4.00)
Jev	16,770	0	0.79	0.88	+0.08	0.08	0.78	+0.55	0.756	0.207	0.45
Liquid d1	16,770	0	0.72	0.81	+0.09	0.09	0.67	+0.40	0.669	0.268	0.72
GPT-6 Luna Decisions	16,758	12	0.73	0.87	+0.14	0.14	0.77	+0.45	0.701	0.256	0.49

`results/metatool_position/2026-10-11_01`, test files `similar_tools` and `scenario`, every list

MMLU: the 4 choices of the question

The 4 choices of the question. Each example is sent with the 4 cyclic permutations of the list, so each option is at each position once. 1,000 examples, 4,000 lists. Options in a list: 4. Correct options in a list: 1.

Instruction: "Choose the correct answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 2.00)	Mean probability at each position	Predictions at each position
Jev	4,000	0	0.92	0.94	+0.02	0.04	0.89	+0.77	0.878	0.104	0.26	0.25, 0.25, 0.25, 0.25	1,000, 1,001, 995, 1,004
Liquid d1	4,000	0	0.84	0.89	+0.05	0.05	0.81	+0.65	0.807	0.153	0.42	0.24, 0.25, 0.25, 0.25	971, 1,011, 1,022, 996
GPT-6 Luna Decisions	3,989	11	0.83	0.88	+0.04	0.05	0.80	+0.64	0.796	0.157	0.44	0.26, 0.25, 0.25, 0.24	1,060, 989, 998, 942

`results/mmlu_own/2026-10-11_01`, every list

ECtHR: cases with 1 violated article

The 10 articles. Each example is sent with the 10 cyclic permutations of the list, so each option is at each position once. 655 examples, 6,550 lists. Options in a list: 10. Correct options in a list: 1.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.32)
Jev	6,550	0	0.89	0.86	-0.03	0.03	0.75	+0.66	0.818	0.156	0.52
Liquid d1	6,550	0	0.88	0.87	-0.01	0.01	0.77	+0.66	0.819	0.155	0.51
GPT-6 Luna Decisions	6,550	0	0.90	0.89	0.00	0.01	0.82	+0.72	0.845	0.121	0.47

`results/ecthr_own/2026-10-11_01`, cases with 1 label, every list

ECtHR: the 10 articles and NOTA, cases with 1 violated article

The 10 articles in the released order with NOTA as the last option. 655 examples, 655 lists. Options in a list: 11. Correct options in a list: 1.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.46)
Jev	655	0	0.84	0.74	-0.10	0.11	0.54	+0.44	0.682	0.242	0.95
Liquid d1	655	0	0.88	0.86	-0.03	0.04	0.76	+0.65	0.800	0.154	0.62
GPT-6 Luna Decisions	655	0	0.87	0.83	-0.05	0.05	0.72	+0.63	0.778	0.153	0.76

`results/ecthr_with_none/2026-10-11_01`, cases with 1 label, list `own`

5. Dependence on the list: a correct option is removed

MetaTool: the tool list without one of its two correct tools

The tool list of the example in the released order without one correct tool. Each example gives 2 lists, one per removed tool. 497 examples, 994 lists. Options in a list: 9. Correct options in a list: 1.

Instruction: "Choose every tool in the list that is needed to solve the user's query."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.17)
Jev	994	0	0.86	0.90	+0.04	0.04	0.81	+0.66	0.821	0.163	0.36
Liquid d1	994	0	0.78	0.85	+0.07	0.07	0.72	+0.50	0.732	0.237	0.57
GPT-6 Luna Decisions	987	7	0.67	0.82	+0.15	0.15	0.70	+0.32	0.620	0.303	0.80

`results/metatool_removal/2026-10-11_01`, lists `without_1` and `without_2`

HotpotQA: the passages without one supporting passage

The passages of the question in the released order without one supporting passage. Each question gives one list per removed passage. 7,384 examples, 14,768 lists. Options in a list: 2 to 9. Correct options in a list: 1.

Instruction: "Choose every passage in the list that supports the answer to the question."

Model	Lists answered	Lists refused	Accuracy	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 1.00 to 3.17)
Jev	14,768	0	0.84	0.83	-0.02	0.03	0.72	+0.59	0.758	0.169	0.72
Liquid d1	14,768	0	0.84	0.85	+0.01	0.02	0.76	+0.61	0.778	0.165	0.61
GPT-6 Luna Decisions	14,327	441	0.80	0.80	0.00	0.03	0.69	+0.51	0.703	0.194	0.92

`results/hotpotqa_removal/2026-10-11_01`, lists `without_1` and `without_2`

ECtHR: the articles without one violated article

The 10 articles in the released order without one violated article, for the cases with 2 to 4 violated articles. Each case gives one list per removed article. 192 examples, 435 lists. Options in a list: 9. Correct options in a list: 1 to 3.

Instruction: "Choose every article in the list that the European Court of Human Rights found violated in this case."

Model	Lists answered	Lists refused	Accuracy	Exact match	Mean confidence	Overconfidence	ECE	Prediction minus runner-up	Highest correct minus lowest correct	Highest correct minus highest distractor	Lowest correct minus highest distractor	Mean probability of each correct option, from the highest	Mean probability of the highest distractor	Mean entropy in bits (maximum 3.17)
Jev	435	0	0.90	0.86	0.83	-0.07	0.07	0.70	+0.19	+0.66	+0.47	highest 0.785, lowest 0.593	0.126	0.64
Liquid d1	435	0	0.90	0.81	0.83	-0.07	0.07	0.71	+0.19	+0.65	+0.46	highest 0.779, lowest 0.585	0.130	0.70
GPT-6 Luna Decisions	426	9	0.92	0.85	0.84	-0.08	0.08	0.75	+0.23	+0.71	+0.48	highest 0.805, lowest 0.576	0.096	0.75

`results/ecthr_removal/2026-10-11_01`, lists `without_1` to `without_4`

MetaTool: the second correct option before and after the removal

The second correct option of an example is the correct option with the second-highest probability in the full list.

Model	Examples with both lists answered	Mean probability of the second correct option in the full list	Its mean probability in the list without the correct option that has the highest probability
Jev	497	0.11	0.71
Liquid d1	497	0.16	0.62
GPT-6 Luna Decisions	490	0.11	0.53

`results/metatool_removal/2026-10-11_01`, list `full` and the list without the correct option that has the highest probability there

HotpotQA: the second correct option before and after the removal

The second correct option of an example is the correct option with the second-highest probability in the full list.

Model	Examples with both lists answered	Mean probability of the second correct option in the full list	Its mean probability in the list without the correct option that has the highest probability
Jev	7,384	0.17	0.66
Liquid d1	7,384	0.19	0.68
GPT-6 Luna Decisions	7,044	0.16	0.60

`results/hotpotqa_removal/2026-10-11_01`, list `full` and the list without the correct option that has the highest probability there

ECtHR: the second correct option before and after the removal

The second correct option of an example is the correct option with the second-highest probability in the full list.

Model	Examples with both lists answered	Mean probability of the second correct option in the full list	Its mean probability in the list without the correct option that has the highest probability
Jev	192	0.19	0.71
Liquid d1	192	0.16	0.68
GPT-6 Luna Decisions	183	0.15	0.68

`results/ecthr_removal/2026-10-11_01`, list `full` and the list without the correct option that has the highest probability there

HotpotQA: a yes/no question per passage

Each passage of a question is asked about in a yes/no question of its own. The 10 questions of an example are sent in one request.

Instruction: "Does this passage support the answer to the question? Passage: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	4,955	0	0.45	396 of 1,000	0.09	131 of 3,955
Liquid d1	4,955	0	0.40	375 of 1,000	0.06	125 of 3,955
GPT-6 Luna Decisions	4,951	4	0.52	508 of 1,000	0.06	181 of 3,951

`results/hotpotqa_yes_no/2026-10-11_01`, every question

ECtHR: a yes/no question per article

Each of the 10 articles is asked about in a yes/no question of its own. The 10 questions of a case are sent in one request.

Instruction: "Did the European Court of Human Rights find a violation of this article in this case? Article: <name of the option>: <text of the option>"

Model	Questions answered	Questions refused	Correct options: mean probability of yes	Correct options with yes at 0.50 or more	Distractors: mean probability of yes	Distractors with yes at 0.50 or more
Jev	10,000	0	0.67	964 of 1,090	0.15	607 of 8,910
Liquid d1	10,000	0	0.89	1,050 of 1,090	0.12	790 of 8,910
GPT-6 Luna Decisions	9,886	114	0.74	818 of 1,018	0.07	374 of 8,868

`results/ecthr_yes_no/2026-10-11_01`, every question